Assignment 4: Probabilistic Modeling

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Question 1

A multivariate Gaussian is not an appropriate model for the given dataset of images because:

1. Gaussian models assume continuous data whereas the the pixels that compose an image are discrete and in particular are binary outcomes (0 or 1);
2. generally, pixels within an image are not symmetrically organized; Gaussian are known to work well when a symmetry is present between data, i.e. when data are symmetrically distributed around a mean value depending on a certain variance;
3. Gaussian models cannot understand a spatial information between pixels. For example, pixels in the 1-dimensional image vector $\mathbf{x}^{(n)}$ are organized in a way that does not reflect their general 2-dimensional structure. For this reason a Gaussian model is not the best choice to learn structural relationships.
4. Gaussian models assume a linear dependence between inputs and targets of the form $y = x^T\theta + \epsilon$, where $\epsilon \approx N(\mu, \sigma)$. In our data this relationship is not present.

Question 2

The likelihood function of $\mathbf{p}$ given the dataset is the product of the probabilities of observing each image x_i given the parameter vector $\mathbf{p}$. Specifically:

$$L(\mathbf{p}, \mathbf{x}) = \prod_{i=1}^{N}\prod_{d=1}^{D} p_d^{x_{i,d}}(1-p_d)^{1-x_{i,d}}$$

Question 3

We first define the log-likelihood as

$$l(\mathbf{p}, \mathbf{x}) = \log(L(\mathbf{p}, \mathbf{x}))$$

Since $\log AB = \log A + \log B$, it becomes:

$$\begin{aligned} l(\mathbf{p}, \mathbf{x}) &= \sum_{i=1}^{N}\sum_{d=1}^{D} \log\left(p_d^{x_{i,d}}(1-p_d)^{1-x_{i,d}}\right) \\ &= \sum_{i=1}^{N}\sum_{d=1}^{D} \left[x_{i,d}\log p_d + (1-x_{i,d})\log(1-p_d)\right] \end{aligned}$$

Question 4

The maximum-likelihood estimator for $\mathbf{p}$ is given by

$$\hat{\mathbf{p}}_{\text{MLE}} = \underset{\mathbf{p}}{\operatorname{argmax}}\, l(\mathbf{p}, \mathbf{x})$$

We can find it solving

$$\frac{\partial l(\mathbf{p}, \mathbf{x})}{\partial \mathbf{p}} = 0$$

Since the derivative is a linear operator, we can put it within the sums. We have that

$$\frac{\partial l(\mathbf{p}, \mathbf{x})}{\partial \mathbf{p}} = \sum_{i=1}^{N} \sum_{d=1}^{D} \frac{\partial}{\partial p_d} \left(x_{i,d} \log p_d + (1 - x_{i,d}) \log(1 - p_d) \right)$$

Setting it to zero we get

$$\sum_{i=1}^{N} \sum_{d=1}^{D} \left[\frac{x_{i,d}}{p_d} - \frac{1 - x_{i,d}}{1 - p_d} \right] = 0$$

We multiply both sides by $p_d(1 - p_d)$:

$$\sum_{i=1}^{N} \sum_{d=1}^{D} \left[(1 - p_d)x_{i,d} - p_d(1 - x_{i,d}) \right] = 0$$

Solving for p_d we get the result:

$$\hat{\mathbf{p}}_{\text{MLE}} = \mathbf{X}$$

where

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \dots \\ \mathbf{x}_N \end{bmatrix}$$

is a $N \times D$ matrix.

Question 5

Taking into account the Bayes' theorem

$$P(\mathbf{p}|\mathbf{x}) = \frac{P(\mathbf{x}|\mathbf{p})P(\mathbf{p})}{P(\mathbf{x})}$$

we want to find the optimal configuration for $\mathbf{p}$ that maximizes the posterior distribution $P(\mathbf{p}|\mathbf{x})$. As far as the maximization is concerned we can ignore $P(\mathbf{x})$, because

$$\begin{aligned} P(\mathbf{p}|\mathbf{x}) &\propto P(\mathbf{x}|\mathbf{p})P(\mathbf{p}) \\ &= \prod_{i=1}^{N} \prod_{d=1}^{D} p_d^{x_{i,d}+\alpha-1} (1 - p_d)^{\beta - x_{i,d}} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \end{aligned}$$

As we did for the maximum likelihood estimation, we maximize the log posterior with respect to $\mathbf{p}$.

$$\log P(\mathbf{p}|\mathbf{x}) = \log P(\mathbf{x}|\mathbf{p})P(\mathbf{p}) = \log P(\mathbf{x}|\mathbf{p}) + \log P(\mathbf{p})$$

$\log P(\mathbf{x}|\mathbf{p})$ has been already computed, thus:

$$\frac{\partial \log P(\mathbf{x}|\mathbf{p})}{\partial p_j} = \sum_{i=1}^{N}\sum_{d=1}^{D}\frac{\partial}{\partial p_j}\left(x_{i,d}\log p_d + (1-x_{i,d})\log(1-p_d)\right) = \sum_{i=1}^{N}\left(\frac{x_{i,j}}{p_j} - \frac{1-x_{i,j}}{1-p_j}\right)$$

As far as $\log P(\mathbf{p})$ is concerned instead:

$$\begin{aligned}\log P(\mathbf{p}) &= \log\left(\prod_{d=1}^{D}\frac{1}{B(\alpha,\beta)}p_d^{\alpha-1}(1-p_d)^{\beta-1}\right)\\ &= \sum_{d=1}^{D}\left[-\log B(\alpha,\beta) + (\alpha-1)\log p_d + (\beta-1)\log(1-p_d)\right]\end{aligned}$$

Therefore:

$$\begin{aligned}\frac{\partial \log P(\mathbf{p})}{\partial p_j} &= \sum_{d=1}^{D}\frac{\partial}{\partial p_j}\left[-\log B(\alpha,\beta) + (\alpha-1)\log p_d + (\beta-1)\log(1-p_d)\right]\\ &= \frac{\alpha-1}{p_j} - \frac{\beta-1}{1-p_j}\end{aligned}$$

Combining the two results, we get that:

$$\begin{aligned}\frac{\partial \log P(\mathbf{p}|\mathbf{x})}{\partial p_j} &= \sum_{i=1}^{N}\left(\frac{x_{i,j}}{p_j} - \frac{1-x_{i,j}}{1-p_j}\right) + \left(\frac{\alpha-1}{p_j} - \frac{\beta-1}{1-p_j}\right)\\ &= \frac{\sum_{i=1}^{N}x_{i,j}}{p_j(1-p_j)} - \frac{N}{1-p_j} + \frac{\alpha-1}{p_j} - \frac{\beta-1}{1-p_j}\end{aligned}$$

Setting it to zero:

$$\sum_{i=1}^{N}x_{i,j} - p_jN + (\alpha-1)(1-p_j) - p_j(\beta-1) = 0$$

that becomes

$$p_j(N+\alpha+\beta-2) = \sum_{i=1}^{N}x_{i,j} + \alpha - 1$$

Finally,

$$p_j = \frac{\sum_{i=1}^{N}x_{i,j} + \alpha - 1}{N+\alpha+\beta-2}$$

and

$$\hat{\mathbf{p}}_{\text{MAP}} = \begin{bmatrix}p_1\\ \dots\\ p_D\end{bmatrix}$$